EDUTRACK AI

AI-BASED ACADEMIC RISK DETECTION & SMART LEARNING RECOMMENDATION SYSTEM

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INTRODUCTION

- In modern education systems, large volumes of student academic data are generated every semester. However, most institutions rely on manual analysis or basic result processing methods.
- This project introduces an AI-based system that:
- Analyzes student academic performance intelligently
- Detects weak subjects using predictive logic
- Classifies academic risk levels
- Recommends suitable learning resources
- The focus is on **predictive academic analytics**, not just scoring marks

PROBLEM STATEMENT

1.TRADITIONAL SYSTEMS ONLY CALCULATE TOTAL AND AVERAGE MARKS

2.WEAK SUBJECTS ARE IDENTIFIED USING FIXED PASS/FAIL RULES

3.NO PREDICTION OF FUTURE ACADEMIC PERFORMANCE

4.NO PERSONALIZED LEARNING RECOMMENDATIONS

5.MANUAL MONITORING IS TIME-CONSUMING AND INEFFICIENT

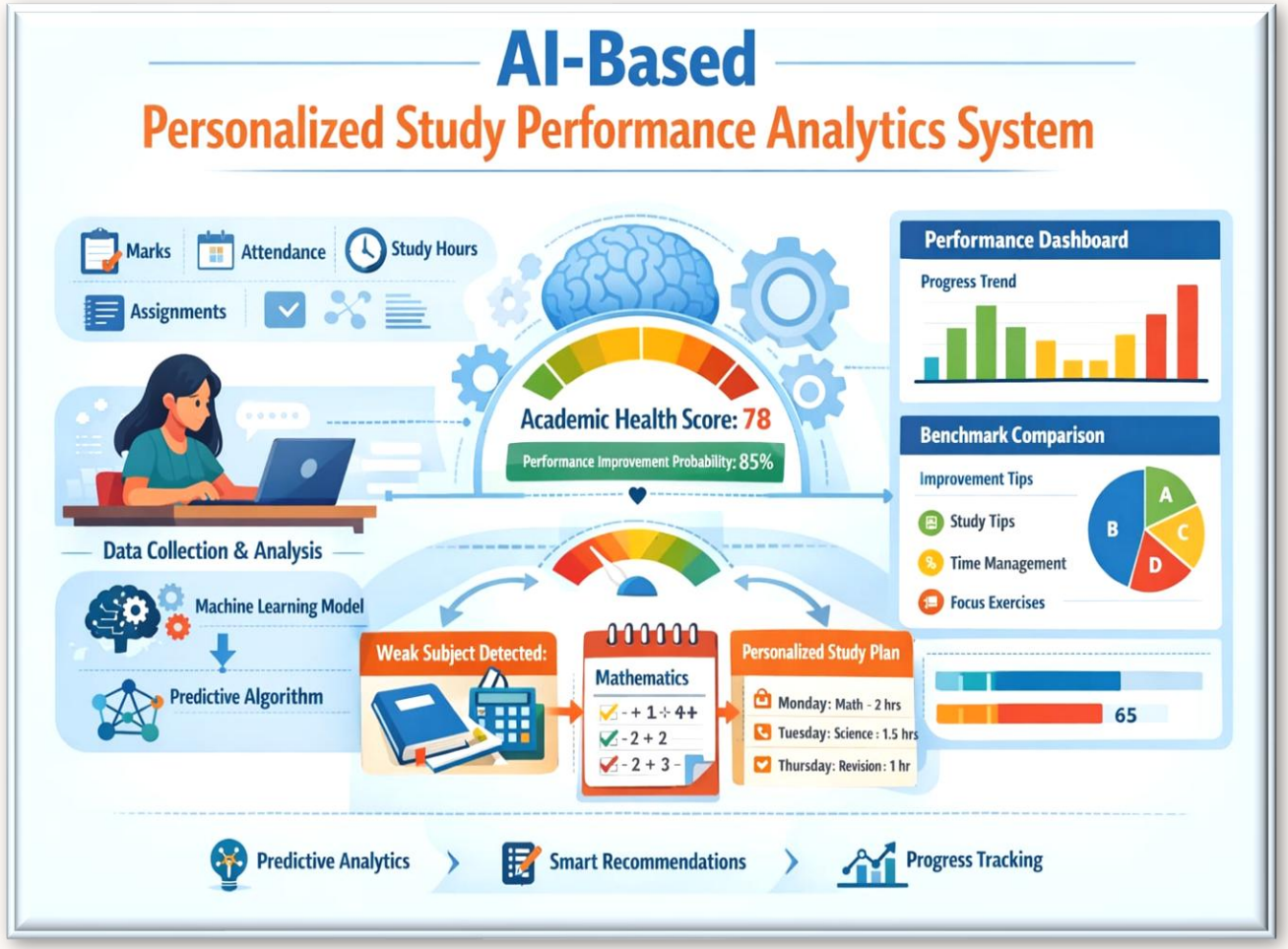
6.THERE IS A NEED FOR A SMART SYSTEM THAT PERFORMS AUTOMATED ACADEMIC RISK DETECTION.

OBJECTIVES

- To develop an **AI-Based Academic Performance Analysis System**.
- To design a **Dynamic Performance Index** using weighted academic parameters.
- To implement **Predictive Risk Detection** for identifying weak subjects.
- To apply **Machine Learning Models** to forecast future performance.
- To perform **Trend Analysis** for monitoring semester-wise improvement or decline.
- To provide **Personalized Learning Recommendations** based on academic gaps.
- To generate **Automated Dashboards & Decision-Support Reports** for data-driven academic monitoring.

PROPOSED SYSTEM

- Develop an **AI-based system** to analyze student academic performance.
- Automatically **detect weak subjects** using smart threshold logic.
- Use **machine learning models** to predict future performance.
- Monitor **semester-wise performance trends** to track improvement or decline.
- Provide **personalized learning recommendations** based on weak subjects.
- Generate an **automated dashboard and performance report** for students and teachers.



TOOLS & TECHNIQUES

- ✓ **Programming Language:**
- **Python**
- ✓ **Machine Learning Library:**
- **Scikit-learn**
- ✓ **Database:**
- **MySQL**
- ✓ **Techniques Used:**
- Regression Analysis
- Decision Tree Classification
- Statistical Threshold Detection
- Trend Analysis
- Rule-Based Recommendation System

EXPECTED OUTCOME

BENEFITS,FUTURE SCOPE,APPLICATIONS

- **Benefits**
  - Early detection of weak subjects
  - Predictive academic performance analysis
  - Personalized learning recommendations
  - Data-driven academic decision making

- **Future Scope**
  - Real-time analytics dashboard
  - Mobile application integration
  - Cloud-based deployment
  - Deep learning-based prediction model
  - Integration with attendance data

- **Applications**
  - Schools
  - Colleges
  - Universities
  - Online learning platforms
  - Educational analytics systems

CONCLUSION

The **AI-Based Academic Risk Detection & Smart Learning Recommendation System** transforms traditional academic evaluation into an **intelligent and data-driven monitoring framework.** Instead of simply calculating marks, the system performs **advanced performance analysis**, identifies **weak subjects through predictive risk detection**, and forecasts **future academic outcomes** using machine learning techniques. By integrating **Predictive Analytics and AI algorithms**, the project enables **early academic risk identification** and supports timely intervention strategies. Overall, the system serves as a **Smart Academic Decision-Support Platform** that enhances student performance, reduces manual workload, and promotes **data-driven educational management and strategic academic planning.**

THANK YOU